# Temporal Scaffolding and Causal World-Models in Narrative Comprehension

*Participant Task Battery — Implementation Documentation*

Version 0.1.0-dev

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## Contents

Part 1 — Task content. Stimuli and items exactly as presented to participants: reading orders, comprehension items, event-card labels, counterfactual probes, and operational constants.

Part 2 — Justifications and audit notes. Design rationale for every non-obvious decision, subtle-cue audit records, alternatives considered, and tracked cleanup items for later sessions.

Part 3 — Implementation notes. Deployment-facing operational details: jsPsych project structure, data schema summary, Prolific and Cognition integration, deployment checklist.

# Part 1 — Task Content

This part contains the content presented to participants, verbatim. Design rationale and audit trails are in Part 2.

## 1.1 Per-story task sequence

Each story is presented with the following task sequence, in fixed order:

- Story reading (minimum 60 seconds, untimed otherwise)
- Comprehension check (6 items, Yes / No / Unsure, randomised order)
- Chronological ordering (drag 8 event cards from a fixed scrambled starting order, plus a 1–7 confidence slider)
- Directed pair scaling (56 ordered pairs, one per screen, 0–6 scale)
- Counterfactual probes (8 probes, 5-point signed scale)

Participants are shown four of the eight stories per session, selected via a preregistered 32-row Latin-square assignment table (see §1.2).

## 1.2 Preregistered constants

### Fixed scrambled starting order (chronological ordering task)

The chronological ordering task begins from the same scrambled sequence across all participants, stories, and conditions: `(E2, E4, E5, E8, E6, E7, E3, E1)`. This sequence was selected to satisfy five preregistered constraints: (i) Kendall τ distance from canonical chronological order E1–E8 equal to 14/28; (ii) Kendall τ distance from the nonlinear reading order (C-B-A-D = E5, E6, E3, E4, E1, E2, E7, E8) equal to 14/28 — ensuring symmetric measurement across conditions; (iii) derangement (no event at its canonical position); (iv) no block-intact adjacencies (no {E1,E2}, {E3,E4}, {E5,E6}, {E7,E8} pair appearing in adjacent positions); (v) no ascending run of three or more canonical-consecutive events at consecutive positions.

### Condition-assignment scheme

Participants are assigned to one of three between-subjects conditions (linear, nonlinear, atemporal) in blocks of 3, with within-block permutation seeded by the participant's Prolific ID. This guarantees exact balance at every multiple of 3 participants, while making condition unpredictable from the participant's perspective.

### Counterbalancing table (32 assignments)

Participants receive one of 32 preregistered story-assignment rows, cycled by participant index. The table is constructed from 4 canonical splits × 2 halves × 4 Williams'-square orderings, subject to the constraint that hospital_incident and care_home_incident (structural replicates) never co-occur in a single assignment.

## 1.3 Event-card labels

These short labels appear on draggable cards in the chronological ordering task and on the source/target cards in the pair-scaling and counterfactual tasks. Each label is ~6–12 words. Audited against Research Notes §5 for subtle-cue neutrality.

### Hospital Incident — Convergent

| ID | Event label (card text) |
|---|---|
| E1 | The hospital administrator announces reduced overnight staffing levels |
| E2 | A contractor disables the ventilator alarm during maintenance |
| E3 | The contractor leaves with the alarm still switched off |
| E4 | The nurse covers a longer patient list than usual |
| E5 | A ventilator tube partially detaches from a patient's mask |
| E6 | The ventilator continues running and no alarm sounds |
| E7 | The nurse finds the patient in respiratory distress |
| E8 | The hospital convenes a formal inquest |

### Care Home Incident — Convergent (structural replicate)

| ID | Event label (card text) |
|---|---|
| E1 | The manager announces reduced overnight staffing |
| E2 | A visiting GP lowers a resident's sedative dose |
| E3 | The medication chart is not updated |
| E4 | The care assistant gives the dose written on the chart |
| E5 | The resident's breathing slows during the night |
| E6 | The care assistant reaches the resident's room after checking larger rooms first |
| E7 | The care assistant finds the resident unresponsive; emergency services are called |
| E8 | The manager convenes a meeting with a regional inspector |

### Community Fair — Three-thread convergence

| ID | Event label (card text) |
|---|---|
| E1 | The organiser books an outdoor venue for the community fair |
| E2 | A first-aid team is assigned to the event |

| ID | Event label (card text) |
|---|---|
| E3 | The first-aid team is called away to another incident |
| E4 | A replacement first-aid team takes nearly two hours to arrive |
| E5 | The temperature by midday is higher than forecast |
| E6 | The organiser opens the rest area and hands out water |
| E7 | An elderly man becomes unresponsive in the sun |
| E8 | The council convenes a review of the event |

## Restaurant Fire — Long chain

| ID | Event label (card text) |
|---|---|
| E1 | A health inspector flags the ventilation as substandard |
| E2 | The owner calls the first available technician for repairs |
| E3 | The technician replaces the fan but not the old ducts |
| E4 | The sous chef notices a faint smell near the back burner |
| E5 | The kitchen is fully booked with all burners running |
| E6 | A cook lights the back burner and the wall ignites |
| E7 | Smoke spreads into the dining room and the restaurant is evacuated |
| E8 | A fire investigation examines the sequence of events |

## School Trip — Fan-out

| ID | Event label (card text) |
|---|---|
| E1 | The teacher chooses a mountain trail for the trip |
| E2 | The group sets out with some students in light jackets |
| E3 | Heavy rain arrives on the ridge |
| E4 | A student loses footing on a wet rock and injures her ankle |
| E5 | The group has no phone signal in the valley |
| E6 | The teacher splits the group to send two students for help |
| E7 | A student under the trees shows signs of hypothermia |
| E8 | A review board examines the teacher's decisions |

## Family Conflict — Chain with feedback

| ID | Event label (card text) |
|---|---|
| E1 | The narrator's father is diagnosed with early-stage dementia |
| E2 | The narrator moves closer to home and begins managing daily care |
| E3 | The sister confirms weekend visits on Thursdays and cancels on Saturdays |
| E4 | The narrator and sister argue; she says he is being controlling |
| E5 | The father's doctor recommends a medication change |
| E6 | The narrator authorises the change without consulting the sister |
| E7 | The sister finds out about the change and calls the narrator |
| E8 | The siblings meet weeks later at a cafe |

## Power Cut — Chain with distal enabler

| ID | Event label (card text) |
|---|---|
| E1 | The council defers tree maintenance along the lane |
| E2 | A neighbour begins using a home oxygen concentrator |
| E3 | An autumn storm brings heavy overnight winds |
| E4 | A branch falls across the power line |
| E5 | The local transformer trips and the street loses power |
| E6 | The neighbour's oxygen concentrator stops running |
| E7 | The narrator finds the neighbour unresponsive and calls emergency services |
| E8 | The council convenes an inquiry |

## Missed Flight — Hourglass

| ID | Event label (card text) |
|---|---|
| E1 | A truck jackknifes on the highway to the airport |
| E2 | The airline's check-in system goes offline for maintenance |
| E3 | The traveller leaves home later than planned |
| E4 | The traveller gets stuck in traffic near the airport |
| E5 | The check-in system comes back online after the flight window closes |
| E6 | The traveller misses the flight |

| ID | Event label (card text) |
| --- | --- |
| E7 | The traveller calls the client from the airport to explain |
| E8 | The client signs with a competitor the following week |

## 1.4 Comprehension check items

Each story has 6 yes/no/unsure items. Each story's item set covers all 8 events, uses 3 compound items (each covering 2 events), 1 single-event item, 1 negative-recall item, and 1 distractor. Every story is balanced 4 Yes / 2 No. All wording has been audited against Research Notes §5.

*Response-time flags: responses faster than 1500ms are tagged* `rushed_flag=true`*; responses slower than 30000ms are tagged* `slow_flag=true`*. Both are annotation-only — neither excludes the response from analysis.*

### Hospital Incident

| # | Item text | Events | Role | Correct |
|---|---|---|---|---|
| 1. | Was a new overnight staffing policy announced, and was the nurse's patient list longer than usual during her night shift? | E1, E4 | compound | Yes |
| 2. | Did the contractor disable the ventilator alarm to run diagnostics? | E2 | single | Yes |
| 3. | Did the contractor switch the alarm back on before leaving the hospital? | E3 | negative recall | No |
| 4. | Did the ventilator tube partially detach at the mask end, with no alarm sounding from the machine? | E5, E6 | compound | Yes |
| 5. | Was the patient stabilised with manual ventilation, and was a formal inquest convened? | E7, E8 | compound | Yes |
| 6. | Did the patient experience a heart attack? | — | distractor | No |

### Care Home Incident

| # | Item text | Events | Role | Correct |
|---|---|---|---|---|
| 1. | Was the manager cutting back on overnight staffing, and was the care assistant covering rooms that had previously been shared? | E1, E4 | compound | Yes |
| 2. | Did the visiting GP lower the sedative dose for one of the residents? | E2 | single | Yes |
| 3. | Did the GP update the medication chart before leaving? | E3 | negative recall | No |
| 4. | Did the resident's breathing slow during the night, with the care assistant reaching her room only after checking the larger rooms first? | E5, E6 | compound | Yes |
| 5. | Was the resident stabilised at the hospital, and did the manager call a meeting with a regional inspector? | E7, E8 | compound | Yes |

| # | Item text | Events | Role | Correct |
|---|---|---|---|---|
| 6. | Did a fire alarm interrupt the evening medication round? | — | distractor | No |

## Community Fair

| # | Item text | Events | Role | Correct |
|---|---|---|---|---|
| 1. | Did the council approve the venue permit in early June, and was a first-aid team assigned by the volunteer coordination office for the day of the fair? | E1, E2 | compound | Yes |
| 2. | Was the first-aid team called away to another incident, and did their replacement take nearly two hours to arrive? | E3, E4 | compound | Yes |
| 3. | Was the temperature by midday well above what had been forecast, and did the organiser open the rest area earlier than planned? | E5, E6 | compound | Yes |
| 4. | Was a first-aid team present on site when the organiser called an ambulance for the elderly man? | E7 | negative recall | No |
| 5. | Did the council convene a review of the venue booking, volunteer scheduling, and weather conditions? | E8 | single | Yes |
| 6. | Was the fair cancelled due to rain? | — | distractor | No |

## Restaurant Fire

| # | Item text | Events | Role | Correct |
|---|---|---|---|---|
| 1. | Did a health inspector tell the owner the ventilation system was not up to standard, and did the owner call the first available technician rather than the original installer? | E1, E2 | compound | Yes |
| 2. | Did the technician replace the old ducts behind the walls? | E3 | negative recall | No |
| 3. | Did the sous chef notice a faint smell near the back burner within hours of the technician's visit? | E4 | single | Yes |
| 4. | Was the kitchen fully booked with all six burners running when a low whump came from inside the wall after a line cook lit the back burner? | E5, E6 | compound | Yes |

| # | Item text | Events | Role | Correct |
|---|---|---|---|---|
| 5. | Did smoke spread into the dining room through the ceiling, and did a fire investigation examine the technician's work order? | E7, E8 | compound | Yes |
| 6. | Did the restaurant's sprinkler system activate during the fire? | — | distractor | No |

## School Trip

| # | Item text | Events | Role | Correct |
|---|---|---|---|---|
| 1. | Did the teacher choose a mountain trail that looped through a wooded section, and did she note that several students were wearing light jackets rather than waterproofs? | E1, E2 | compound | Yes |
| 2. | Did the teacher decide to turn the group back at the high point? | E3 | negative recall | No |
| 3. | Did a student injure her ankle after losing footing on a wet rock? | E4 | single | Yes |
| 4. | Was there no phone signal in the valley, and did the teacher send two older students back toward the car park to call for rescue? | E5, E6 | compound | Yes |
| 5. | Did one of the younger students show early signs of hypothermia under the trees, and did a review board examine the teacher's decisions the following week? | E7, E8 | compound | Yes |
| 6. | Did a member of the group suffer a concussion from the fall? | — | distractor | No |

## Family Conflict

| # | Item text | Events | Role | Correct |
|---|---|---|---|---|
| 1. | Was the narrator's father diagnosed with early-stage dementia, and did the narrator move closer to home that summer? | E1, E2 | compound | Yes |
| 2. | Did the sister come every weekend through the autumn? | E3 | negative recall | No |
| 3. | Did the sister accuse the narrator of being controlling during a phone call? | E4 | single | Yes |

| # | Item text | Events | Role | Correct |
|---|---|---|---|---|
| 4. | Did the father's doctor recommend changing his medication, and did the narrator sign the form for the new prescription without calling the sister? | E5, E6 | compound | Yes |
| 5. | Did the sister find out about the medication change from a neighbour, and did the siblings meet at a cafe a few weeks later? | E7, E8 | compound | Yes |
| 6. | Did the narrator consider moving their father into a nursing home? | — | distractor | No |

## Power Cut

| # | Item text | Events | Role | Correct |
|---|---|---|---|---|
| 1. | Had the council deferred tree maintenance near the power lines for over a year? | E1 | single | Yes |
| 2. | Did the healthcare provider supply a backup battery with the oxygen concentrator? | E2 | negative recall | No |
| 3. | Did the forecast warn of heavy winds overnight, and did a branch come down across the power line during the storm? | E3, E4 | compound | Yes |
| 4. | Did the local transformer trip, and did the neighbour's oxygen machine shut down when the power went out? | E5, E6 | compound | Yes |
| 5. | Did the narrator find the neighbour unresponsive, and did a council inquiry examine the budget meetings where tree work had been deferred? | E7, E8 | compound | Yes |
| 6. | Did the power company issue an apology to the street after the outage? | — | distractor | No |

## Missed Flight

| # | Item text | Events | Role | Correct |
|---|---|---|---|---|
| 1. | Did a truck jackknife on the main highway on the morning of the traveller's flight, and had the airline's check-in system gone offline overnight for scheduled maintenance? | E1, E2 | compound | Yes |
| 2. | Did the traveller leave the house later than planned, and did he hit traffic about twenty minutes from the airport? | E3, E4 | compound | Yes |

| # | Item text | Events | Role | Correct |
|---|---|---|---|---|
| 3. | Did the check-in desks reopen before the window for the traveller's flight had closed? | E5 | negative recall | No |
| 4. | Did the traveller miss the flight and call the client from the airport? | E6, E7 | compound | Yes |
| 5. | Did the traveller learn the following week that the client had signed with a competitor? | E8 | single | Yes |
| 6. | Did the traveller manage to catch a later flight that same afternoon? | — | distractor | No |

## 1.5 Counterfactual probes

Each story has 8 probes: 6 anchor probes (one per non-outcome event E1–E6 asking about its effect on E7), 1 sibling null (a pair of events within the story that are genuinely causally independent), and 1 reverse null (E8 → E7, probing an effect-to-cause direction). All probes use a 5-point signed scale: Much less likely / Slightly less likely / No change / Slightly more likely / Much more likely.

E7 serves as the consistent outcome position across all 8 stories; this structural parallelism is deliberate for cross-story aggregate analyses. For missed_flight, E7 ("the traveller calls the client from the airport to explain") is the outcome of interest; E6 ("the traveller misses the flight") functions as the proximate-cause anchor probe and is expected to produce the strongest "less likely" response in healthy discrimination.

### Hospital Incident

| Role | A → C | Probe prompt |
|---|---|---|
| anchor | E1 → E7 | If the hospital administrator had not announced reduced overnight staffing, would the patient still have ended up in respiratory distress? |
| anchor | E2 → E7 | If the contractor had not disabled the ventilator alarm during maintenance, would the patient still have ended up in respiratory distress? |
| anchor | E3 → E7 | If the contractor had switched the alarm back on before leaving, would the patient still have ended up in respiratory distress? |
| anchor | E4 → E7 | If the nurse had not been assigned a longer patient list than usual, would the patient still have ended up in respiratory distress? |
| anchor | E5 → E7 | If the ventilator tube had not partially detached from the patient's mask, would the patient still have ended up in respiratory distress? |
| anchor | E6 → E7 | If an alarm had sounded when the ventilator stopped delivering air, would the patient still have ended up in respiratory distress? |
| sibling null | E1 → E2 | If the hospital administrator had not announced reduced overnight staffing, would the contractor still have disabled the ventilator alarm during maintenance? |
| reverse null | E8 → E7 | If the hospital had not convened a formal inquest, would the patient still have ended up in respiratory distress? |

### Care Home Incident

| Role | A → C | Probe prompt |
|---|---|---|
| anchor | E1 → E7 | If the manager had not announced reduced overnight staffing, would the care assistant still have found the resident unresponsive? |

| Role | A → C | Probe prompt |
|---|---|---|
| anchor | E2 → E7 | If the visiting GP had not lowered the resident's sedative dose, would the care assistant still have found the resident unresponsive? |
| anchor | E3 → E7 | If the medication chart had been updated, would the care assistant still have found the resident unresponsive? |
| anchor | E4 → E7 | If the care assistant had given a different dose than the one written on the chart, would the care assistant still have found the resident unresponsive? |
| anchor | E5 → E7 | If the resident's breathing had not slowed during the night, would the care assistant still have found the resident unresponsive? |
| anchor | E6 → E7 | If the care assistant had reached the resident's room before checking the larger rooms, would the care assistant still have found the resident unresponsive? |
| sibling null | E1 → E2 | If the manager had not announced reduced overnight staffing, would the visiting GP still have lowered the resident's sedative dose? |
| reverse null | E8 → E7 | If the manager had not convened a meeting with a regional inspector, would the care assistant still have found the resident unresponsive? |

## Community Fair

| Role | A → C | Probe prompt |
|---|---|---|
| anchor | E1 → E7 | If the organiser had not booked an outdoor venue for the fair, would the elderly man still have become unresponsive in the sun? |
| anchor | E2 → E7 | If a first-aid team had not been assigned to the event, would the elderly man still have become unresponsive in the sun? |
| anchor | E3 → E7 | If the first-aid team had not been called away to another incident, would the elderly man still have become unresponsive in the sun? |
| anchor | E4 → E7 | If a replacement first-aid team had arrived within a few minutes, would the elderly man still have become unresponsive in the sun? |
| anchor | E5 → E7 | If the temperature by midday had not been higher than forecast, would the elderly man still have become unresponsive in the sun? |
| anchor | E6 → E7 | If the organiser had not opened the rest area and handed out water, would the elderly man still have become unresponsive in the sun? |
| sibling null | E2 → E5 | If a first-aid team had not been assigned to the event, would the temperature by midday still have been higher than forecast? |
| reverse null | E8 → E7 | If the council had not convened a review of the event, would the elderly man still have become unresponsive in the sun? |

## Restaurant Fire

| Role | A → C | Probe prompt |
|---|---|---|
| anchor | E1 → E7 | If the health inspector had not flagged the ventilation as substandard, would smoke still have spread into the dining room and the restaurant been evacuated? |
| anchor | E2 → E7 | If the owner had called the original installer rather than the first available technician, would smoke still have spread into the dining room and the restaurant been evacuated? |
| anchor | E3 → E7 | If the technician had replaced the old ducts along with the fan, would smoke still have spread into the dining room and the restaurant been evacuated? |
| anchor | E4 → E7 | If the sous chef had not noticed a faint smell near the back burner, would smoke still have spread into the dining room and the restaurant been evacuated? |
| anchor | E5 → E7 | If the kitchen had not been fully booked with all burners running, would smoke still have spread into the dining room and the restaurant been evacuated? |
| anchor | E6 → E7 | If the wall had not ignited when a cook lit the back burner, would smoke still have spread into the dining room and the restaurant been evacuated? |
| sibling null | E3 → E5 | If the technician had replaced the old ducts along with the fan, would the kitchen still have been fully booked with all burners running? |
| reverse null | E8 → E7 | If a fire investigation had not examined the sequence of events, would smoke still have spread into the dining room and the restaurant been evacuated? |

## School Trip

| Role | A → C | Probe prompt |
|---|---|---|
| anchor | E1 → E7 | If the teacher had not chosen a mountain trail for the trip, would a student still have shown signs of hypothermia under the trees? |
| anchor | E2 → E7 | If the group had set out with full wet-weather gear for every student, would a student still have shown signs of hypothermia under the trees? |
| anchor | E3 → E7 | If heavy rain had not arrived on the ridge, would a student still have shown signs of hypothermia under the trees? |
| anchor | E4 → E7 | If the student had not lost footing on a wet rock and injured her ankle, would a student still have shown signs of hypothermia under the trees? |

| Role | A → C | Probe prompt |
|---|---|---|
| anchor | E5 → E7 | If the group had kept a phone signal in the valley, would a student still have shown signs of hypothermia under the trees? |
| anchor | E6 → E7 | If the teacher had not split the group to send two students for help, would a student still have shown signs of hypothermia under the trees? |
| sibling null | E2 → E3 | If the group had set out with full wet-weather gear for every student, would heavy rain still have arrived on the ridge? |
| reverse null | E8 → E7 | If a review board had not examined the teacher's decisions, would a student still have shown signs of hypothermia under the trees? |

## Family Conflict

| Role | A → C | Probe prompt |
|---|---|---|
| anchor | E1 → E7 | If the narrator's father had not been diagnosed with early-stage dementia, would the sister still have found out from a neighbour and called the narrator? |
| anchor | E2 → E7 | If the narrator had not moved closer to home to manage daily care, would the sister still have found out from a neighbour and called the narrator? |
| anchor | E3 → E7 | If the sister had kept her weekend visits as arranged, would the sister still have found out from a neighbour and called the narrator? |
| anchor | E4 → E7 | If the narrator and sister had not argued on the phone, would the sister still have found out from a neighbour and called the narrator? |
| anchor | E5 → E7 | If the father's doctor had not recommended a medication change, would the sister still have found out from a neighbour and called the narrator? |
| anchor | E6 → E7 | If the narrator had consulted the sister before authorising the change, would the sister still have found out from a neighbour and called the narrator? |
| sibling null | E1 → E3 | If the narrator's father had not been diagnosed with early-stage dementia, would the sister still have cancelled her weekend visits? |
| reverse null | E8 → E7 | If the siblings had not met weeks later at a cafe, would the sister still have found out from a neighbour and called the narrator? |

## Power Cut

| Role | A → C | Probe prompt |
|---|---|---|
| anchor | E1 → E7 | If the council had kept up tree maintenance along the lane, would |

| Role | A → C | Probe prompt |
|---|---|---|
| | | the narrator still have found the neighbour unresponsive? |
| anchor | E2 → E7 | If the neighbour had not begun using a home oxygen concentrator, would the narrator still have found the neighbour unresponsive? |
| anchor | E3 → E7 | If an autumn storm had not brought heavy overnight winds, would the narrator still have found the neighbour unresponsive? |
| anchor | E4 → E7 | If no branch had fallen across the power line, would the narrator still have found the neighbour unresponsive? |
| anchor | E5 → E7 | If the local transformer had not tripped, would the narrator still have found the neighbour unresponsive? |
| anchor | E6 → E7 | If the neighbour's oxygen concentrator had kept running through the outage, would the narrator still have found the neighbour unresponsive? |
| sibling null | E2 → E3 | If the neighbour had not begun using a home oxygen concentrator, would an autumn storm still have brought heavy overnight winds? |
| reverse null | E8 → E7 | If the council had not convened an inquiry, would the narrator still have found the neighbour unresponsive? |

## Missed Flight

| Role | A → C | Probe prompt |
|---|---|---|
| anchor | E1 → E7 | If a truck had not jackknifed on the highway to the airport, would the traveller still have called the client from the airport to explain? |
| anchor | E2 → E7 | If the airline's check-in system had not gone offline for maintenance, would the traveller still have called the client from the airport to explain? |
| anchor | E3 → E7 | If the traveller had left home earlier than planned, would the traveller still have called the client from the airport to explain? |
| anchor | E4 → E7 | If the traveller had not been stuck in traffic near the airport, would the traveller still have called the client from the airport to explain? |
| anchor | E5 → E7 | If the check-in system had come back online before the flight window closed, would the traveller still have called the client from the airport to explain? |
| anchor | E6 → E7 | If the traveller had not missed the flight, would the traveller still have called the client from the airport to explain? |
| sibling null | E1 → E2 | If a truck had not jackknifed on the highway to the airport, would the airline's check-in system still have gone offline for maintenance? |
| reverse null | E8 → E7 | If the client had not signed with a competitor the following week, |

| Role | A → C | Probe prompt |
|---|---|---|
| | | would the traveller still have called the client from the airport to explain? |

## 1.6 Operational thresholds and flags

| Setting | Value | Meaning |
|---|---|---|
| Minimum reading time | 60,000 ms | Continue button disabled until reached |
| Comprehension rushed flag | < 1,500 ms | Annotation only; does not exclude |
| Comprehension slow flag | > 30,000 ms | Annotation only; does not exclude |
| Pair-scaling rushed flag | < 500 ms | Annotation only; does not exclude |
| Prolific redirect buffer | 2,000 ms | Post-debrief delay for data upload |
| Attention check (IMC) | "select Orange" | Session start; failure flags, does not eject |
| Minimum viewport | 900 × 600 px | Smaller viewport rejected with return-to-Prolific message |
| Mobile detection | User-agent regex | Mobile devices rejected at session start |

# Part 2 — Justifications and Audit Notes

This part documents the rationale for non-obvious design decisions made during this implementation session, including alternatives considered, trade-offs accepted, and items flagged for follow-up.

## 2.1 Scrambled starting order — construction and selection

The sequence `(E2, E4, E5, E8, E6, E7, E3, E1)` was selected from 101 permutations that satisfy all five hard constraints (see Part 1 §1.2). Among those, it was chosen for three reasons:

- Best inversion uniformity. Per-position inversion counts are [1, 2, 2, 4, 2, 2, 1, 0], with standard deviation 1.00 — the lowest among candidates. No single region of the sequence is trivially easy or disproportionately hard to resolve.
- E8 at position 4 (middle). Still an unambiguous wrong position — so the §5.7 manipulation-check logic holds — but without placing the resolution event so early that a participant might doubt their own reading.
- Structural symmetry with canonical. The permutation is roughly a rotational offset of the canonical sequence (E2 where E1 was, ending with E1), which is easy to defend in prose and easy to communicate to reviewers.

### On the τ-symmetry constraint to C-B-A-D

The original task design required $\tau \approx 14$ from canonical only. An additional constraint — $\tau \geq 13$ from C-B-A-D as well — was added because it guarantees neither the linear nor the nonlinear condition starts the ordering task closer to their reading order than the other. Without this, a scramble $\tau=14$ from canonical could land arbitrarily close to C-B-A-D, making differential task difficulty across conditions a plausible alternative explanation for any condition effect observed on the ordering task. The chosen sequence achieves exact $\tau=14$ to both anchors.

## 2.2 Comprehension items — packing strategy and audit

Each story has 6 items; all 8 events must be covered. Single-item-per-event would yield 8 items, exceeding the budget. The adopted pattern is 3 compound items (each covering 2 events) + 2 single-event items (1 positive + 1 negative-recall) + 1 distractor. This fits the 6-item budget, covers all 8 events, and balances item roles parametrically across stories for by-role analysis.

### Subtle-cue audit (Research Notes §5)

Every item was checked against six rules derived from Research Notes §5:

- No causal verbs (because, caused, led to, triggered, resulted in)
- No counterfactual constructions (would have, if only, under normal circumstances)
- No retrospective narrator signals (it turns out, looking back, in hindsight)
- No characterising nouns that pre-package events ("pattern", "failure")
- No normative adjectives ("right", "properly") — neutral forms ("updated") only

- No cross-event temporal connectives ("later", "subsequently") that would be stripped in the atemporal condition, making items condition-sensitive

### Structural replicate parallelism

Hospital and Care Home items are structurally isomorphic: Item 1 maps staffing + workload; Item 2 covers the "maintenance/medical expert action"; Item 3 is a negative-recall "handover failure"; Item 4 covers the climax convergence; Item 5 covers resolution + inquest; Item 6 is a distractor. Preserving this parallelism lets the pilot detect domain-specific confounds: if replicates behave differently on matched items, domain content is contributing beyond pure topology.

## 2.3 Event-card labels — drafting choices

All 64 labels were drafted from scratch from the stimulus text (not reused from author_intended_graphs.json), because the original graph labels contain language that does subtle causal work — appropriate for the design blueprint but inappropriate for participant presentation. Examples of problematic graph labels:

- school_trip E3: "The weather deteriorates faster than forecast" — "faster than" frames the weather as an independent causal pusher
- power_cut E4: "An unmaintained branch falls across a power line" — "unmaintained" causally links E1 to E4 on the card
- restaurant_fire E6: "Grease from an overloaded pan catches fire near the leak" — "near the leak" pre-connects causally

### Three flagged judgement calls

**school_trip E1.** Removed "despite the forecast" from the teacher-chooses-trail sentence. "Despite" does causal work by framing the choice as being made against a warning signal, which would pre-announce the E1 → E3 connection on the card. Counter-consideration: the forecast-awareness is relevant to the E8 review; this information is preserved in the prose but not the card.

**family_conflict E3.** Rewrote the story's "visits became unreliable" into behaviourally explicit wording ("confirms visits on Thursdays and cancels on Saturdays"). §5 flagged exactly this kind of characterising word ("pattern", "unreliable") as the audit target for this story. The behavioural rephrasing tracks the story's own specific sentences.

**power_cut E2.** Dropped "without backup battery" from the oxygen-concentrator card. This detail is critical to the causal chain and appears explicitly in the story, but on a card it reads as a deliberate signal ("this matters"), whereas the prose gives it incidentally. The information is captured in the comprehension item instead. Trade-off: if participants undervalue E2 → E6 in pair scaling, this may be an artefact of thin card content rather than pure representation.

## 2.4 Counterfactual probes — sibling-null selection

Task design §7.1 specifies hospital's sibling null (E1 → E2). The other 7 sibling nulls were chosen to instantiate the same criterion (two genuinely independent events within the story,

typically two causal "roots"). Two choices required judgement because the finalised stimulus set replaced server_outage → care_home_incident and coastal_flooding → community_fair:

**care_home_incident.** E1 → E2 (staffing policy and GP's medical decision). Direct structural analogue of the original server_outage probe: two independent root actions by different actors.

**community_fair.** E2 → E5 (first-aid team assignment and weather). Chose E2 → E5 over E1 → E5 ("if no venue booked, would temperature still exceed forecast?") because undermining the venue booking at all undermines the scenario itself, not just the causal chain.

### Discrimination index (per §7.5)

Each CF summary row includes a discrimination index: mean |anchor − 3| − mean |null − 3|, where 3 is the "no change" midpoint. Positive values indicate the participant distinguishes causal anchors from independent events; near-zero suggests flat responding; negative is diagnostic of confusion or motivated over-rating. This computation is pre-committed at the summary row, not left to analysis-time.

## 2.5 Counterbalancing design

The 32-row assignment table comes from 4 canonical splits × 2 halves × 4 Williams'-square orderings. Williams's square balances first-order carryover — within the table, every ordered pair (X, Y) with X ≠ Y appears as an adjacent pair approximately equally often.

### Hospital / Care Home non-co-occurrence (added this session)

Hospital_incident and care_home_incident are structural replicates: same convergent topology, same theme (medical oversight failure involving an overnight shift and a vulnerable patient), and highly parallel item wording. A participant receiving both would almost certainly (a) carry over causal logic from one to the other, (b) infer that the two are related and form meta-hypotheses about the study, and (c) produce within-subject contamination that would be impossible to untangle at analysis time.

The splits were rebuilt to enforce non-co-occurrence at the design level. All 32 assignments satisfy this constraint. Consequences: carryover coverage drops from 56 to 48 distinct ordered pairs (the 8 pairs involving hospital ↔ care_home are impossible by design). Position balance and story-frequency balance are preserved exactly. The cost is that replicates cannot be compared within-participant, which was never the intended design anyway — they are compared between participants to detect domain-specific confounds.

### Condition-assignment perturbation — small-N drift

The condition-assignment scheme (blocks of 3, permuted by hashed participant ID) produces exact balance at N = 3k for k ∈ ℕ. At small N it can drift slightly — in one simulation of N = 10 it produced 2 linear / 3 nonlinear / 5 atemporal. At realistic collection sizes (N ≥ 96) the imbalance disappears. The trade-off accepted: a small-N imbalance in exchange for condition unpredictability from the participant's perspective. Flag for pilot review: if interim analyses are planned before N = 96, the imbalance should be weighted in those analyses.

## 2.6 Pair-scaling — one-at-a-time versus matrix

Considered alternative: show all 56 pairs as a matrix so participants can consciously calibrate ratings against each other. Rejected for three reasons:

- The primary analysis is Mantel RSA between the human directed matrix and the model's asymmetric RDM. The model produces its RDM pair-by-pair; the human ratings must be generated under roughly matched conditions for the Mantel correlation to measure cross-system representational agreement. A matrix layout shifts humans into deliberate internal calibration, which has no counterpart on the model side.
- Task design §6.3 specifies framing-effect control as an exploratory analysis — each unordered pair rated in both directions. Matrix layout exposes both directions simultaneously, making the framing check uninterpretable.
- Task design §6.2 explicitly specifies one-at-a-time presentation.

## 2.7 Ordering task — SortableJS over SurveyJS ranking

During the Phase 2 scaffold planning the recommendation was SurveyJS's ranking question (via @jspsych/plugin-survey) for a first-class jsPsych 8 component. For the ordering task specifically, this was switched to a custom SortableJS-based trial because:

- SortableJS exposes per-drag onStart / onEnd hooks, enabling fine-grained event logging (which cards the participant moved, null-drops, total drag count). SurveyJS ranking does not expose drag events for post-hoc logging.
- The ordering task is the one place in the battery where per-interaction event detail is genuinely informative — drag behaviour correlates with confidence and can flag surface-level engagement issues.
- The confidence slider is rendered on the same page as the drag list (task design §5.3), which a single custom trial handles cleanly.

## 2.8 Attention check strategy

One session-start instructional manipulation check (IMC) — select "Orange" — is included. Mid-experiment IMCs were considered and rejected: the battery already includes rich domain-specific quality controls (comprehension distractors, reverse-null CF probes, rushed-response flags, minimum reading time, null control probes), and additional IMCs would consume session time without meaningful marginal information. The IMC is configured flag-not-eject by default; failure annotates the data record but does not end the session.

## 2.9 Dual-level logging

The default logging level is "detail", which emits both a flat trial-summary row and a nested events array containing per-action timestamps (scrolls, focus losses, drag starts/ends, slider touches). The alternative "summary" mode strips the events array to reduce file size.

Rationale for detail default: storage cost is trivial (~300 KB per participant in synthetic testing, scaling to ~60 MB at N = 200), the option value is high (enables post-hoc data-quality audits that cannot be retrofitted), and detail can always be collapsed to summary at analysis time. Matches

the project's model-side commitment to preserving raw behaviour and handling equivalences at evaluation time.

## 2.10 Synthetic end-to-end verification

A full-pipeline simulation was run for three synthetic participants (one per condition). Results:

| Metric | Value |
|---|---|
| Rows per participant | 313 (6 intro + 75 × 4 stories + 3 outro + 4 break screens) |
| Universal metadata fields on every row | participant_id, condition, assignment_id, experiment_version |
| Per-story summary rows | 5 (comprehension, ordering, pair-scaling, CF, plus reading) |
| Pair-scaling matrix shape | 8×8 with diagonal = null (8 nulls, 56 off-diagonal) |
| CF anchor vector length | 6 (E1–E6 in canonical order) |
| Hospital / Care Home co-occurrence | 0 / 32 assignments (confirmed at runtime) |
| Scrambled-order τ distances | 14 from canonical, 14 from C-B-A-D (runtime confirmed) |
| Shape issues found | 0 across all three simulated participants |

## 2.11 Items tracked for follow-up

The following items were identified during this session and deferred to separate work:

**Review other story pairs for thematic co-occurrence risks.** Hospital / care_home was the clearest case (structural replicates). Worth a systematic thematic-similarity audit of the remaining pairs before data collection — e.g., the two hourglass/fan-out stories (school_trip, missed_flight) share a "personal decision leading to adverse outcome" structure that may warrant separation, though they are topologically distinct. Decision: flagged, not yet executed.

**Clean up author_intended_graphs.json.** The project's design-scaffold file still lists obsolete entries (server_outage, coastal_flooding) that were replaced by care_home_incident and community_fair in the finalised stimuli. The graph file needs its events and causal_edges reauthored for the two replacement domains. This does not affect data collection but should be resolved before write-up.

**Consent-text placeholders.** The consent screen contains bracketed placeholders: [INSTITUTION], [ETHICS_REF], [RESEARCHER], [EMAIL], [SUPERVISOR], [SUPERVISOR_EMAIL]. These must be replaced with the specific ethics approval details before deployment.

**Prolific completion URL.** CONFIG.prolific.completion_url contains the placeholder cc=REPLACE_WITH_STUDY_CODE; the actual study completion code from Prolific must be substituted per deployed study.

**SurveyJS-ranking fallback path is documented but untested.** If SortableJS misbehaves on Cognition-hosted deployment, the documented fallback (SurveyJS ranking via @jspsych/plugin-survey) can be substituted in tasks/ordering.js. Untested because not needed, but documented as a known mitigation path.

**Stimuli power-analysis flagged elsewhere.** Per the project's ongoing note: probe in a separate chat whether the stimuli are powerful enough to produce an effect. Re-raise during pilot planning or power-analysis work.

**Simulator-only artefact.** The synthetic runner cannot exercise the confidence slider's DOM interaction path, so ordering.confidence_rating appears as null in the simulated output. In real browser runs this populates on first pointerdown/keypress. If null appears in live data, treat it as a real signal (slider never touched), not a simulator quirk.

# Part 3 — Implementation Notes

Operational details for deploying the experiment to Cognition.run, integrating with Prolific, and interpreting the output data. Intended as a deployment reference for future sessions, not a full technical specification.

## 3.1 Tech stack

| Component | Choice |
|---|---|
| Experiment framework | jsPsych 8.2.1 |
| Plugins (CDN) | @jspsych/plugin-html-button-response, -html-slider-response, -html-keyboard-response, -instructions, -preload, -call-function, -survey-html-form |
| Drag-to-reorder | SortableJS 1.15.2 (custom trial using html-button-response as the outer shell) |
| Hosting | Cognition.run |
| Recruitment | Prolific |
| Data format | JSON export preferred (preserves nested events and matrix structure); CSV also supported |

## 3.2 Project layout

Directory structure during development (content and logic kept separate so copy-edits to items never require code changes):

```
experiment/
├── src/
│   ├── config.js                   # versions, task params, preregistered constants,
feature flags
│   ├── main.js                     # timeline entry; glues intro + per-story blocks +
outro
│   ├── condition.js                # between-subjects assignment
│   ├── selection.js                # story-selection from 32-row assignment table
│   ├── data.js                     # participant metadata, Prolific redirect helpers
│   ├── utils.js                    # URL params, timestamps, scroll/focus tracker,
shuffle
│   ├── intro.js                    # browser check, welcome, consent, demographics,
IMC, instructions
│   ├── inter_story.js              # between-story break + progress indicator
│   ├── outro.js                    # comments, debrief, Prolific redirect
│   └── tasks/
│       ├── story_reading.js        # min-time threshold, activity tracking
│       ├── comprehension.js        # 6 items + summary row
│       ├── ordering.js             # SortableJS + confidence slider, one row + detail
events
│       ├── pair_scaling.js         # 56 pairs + 8x8 matrix summary row
│       └── counterfactual.js       # 8 probes + anchor vector + discrimination index
├── stimuli/
│   ├── stories.js                  # 8 stories × 3 versions
(linear/nonlinear/atemporal)
```

```
│   ├── comprehension.js          # 48 comprehension items
│   ├── event_cards.js            # 64 short event labels
│   ├── cf_probes.js              # 64 counterfactual probes
│   └── assignments.js            # 32-row preregistered assignment table
├── assets/
│   └── style.css
├── index.html                    # local test harness — not used on Cognition
├── smoke_test.js                 # Node sandbox verification of all globals and
builders
├── synthetic_run.js              # end-to-end simulated participant runner
└── DATA_SCHEMA.md                # operational reference for analysis code
```

## 3.3 Cognition deployment

Cognition provides a JavaScript-only code editor — all experiment content must be loaded as .js files setting global variables, not as JSON fetched at runtime. The project is structured to match this deployment model: each content file in stimuli/ assigns to a global (STORIES, COMPREHENSION_ITEMS, EVENT_CARDS, CF_PROBES, ASSIGNMENTS), and every source module in src/ is an IIFE attaching one global (Condition, Selection, DataHelpers, etc.).

Plugins are loaded from the jsDelivr CDN in index.html, with versions pinned in config.js for reproducibility. Bumping a plugin version is a one-line change in index.html (and one line in the documented version-pin table in config.js).

### Deployment checklist

- Replace consent-text placeholders in src/intro.js: [INSTITUTION], [ETHICS_REF], [RESEARCHER], [EMAIL], [SUPERVISOR], [SUPERVISOR_EMAIL]
- Set CONFIG.prolific.completion_url in src/config.js to the study-specific Prolific completion URL
- Set CONFIG.debug.enabled = false (it is already false by default)
- Upload all src/*.js and stimuli/*.js files to Cognition as individual scripts in the order given in index.html
- Test the SurveyJS / SortableJS integration on Cognition — if ranking drag misbehaves, confirm the fallback path in tasks/ordering.js
- Verify that data reaches Cognition before Prolific redirect: use the built-in 2000ms buffer, or increase CONFIG.prolific.redirect_buffer_ms if uploads are consistently slow
- Configure Prolific to pass PROLIFIC_PID, STUDY_ID, SESSION_ID as URL parameters (the experiment reads these automatically)
- For strict preregistered cycle control, also pass ?pIndex=N from Prolific orchestration so participants map to canonical assignments 0..31 in order

## 3.4 Data schema (summary)

Full specification in DATA_SCHEMA.md. Summary: 313 rows per participant (6 intro + 75 × 4 stories + 3 outro + 4 inter-story breaks), each row carrying universal metadata fields (participant_id, condition, assignment_id, experiment_version) plus task-specific fields.

The analysis-ready units are the per-story summary rows:

| Summary row | Primary analysis object |
|---|---|
| comprehension_summary | accuracy (data quality gate) |
| ordering | final_order (length-8 permutation) and kendall_tau_to_canonical |
| pair_scaling_summary | directed_matrix (8×8 RSA-ready asymmetric) |
| counterfactual_summary | anchor_vector (length-6) + discrimination_index |

## 3.5 Built-in verification

Two Node.js-runnable verifiers are included in the repo:

`smoke_test.js` — loads all JS files in a sandboxed context and verifies: global presence (CONFIG, STORIES, …), condition balance simulation, story-selection correctness, event-card coverage, comprehension/pair-scaling/CF block shape, Kendall τ distance computations, and hospital/care_home non-co-occurrence at runtime.

`synthetic_run.js` — instantiates one simulated participant per condition, executes the full timeline through minimal jsPsych stubs, writes trial-by-trial data to synthetic_output.json, and verifies shape against DATA_SCHEMA.md. Catches integration issues that component-level tests miss.

Both scripts are suitable for CI or pre-deployment smoke testing. Neither requires a real browser; both complete in under 2 seconds.